Introduction

Classification is a common machine-learning task in which a model predicts a category based on observed features. In this project, I used the SEMMA methodology—Sample, Explore, Modify, Model, and Assess—to build and evaluate models that classify Iris flowers by species.

The Iris dataset is useful for a small, reproducible classification experiment because it contains four numeric measurements and three known species. The objective was to follow the SEMMA process and compare two classification approaches on the same held-out test data.

Dataset & Research Question

Research Question: How well can two simple models (KNN and a decision tree) tell Iris flower types from their sepal and petal measurements?

The dataset contained 150 observations, with four features: sepal length, sepal width, petal length, and petal width. The target variable contained three species: setosa, versicolor, and virginica. Each species had 50 observations.

The notebook loaded the data using scikit-learn's Iris dataset loader. A Kaggle page was used as a dataset reference: Iris Species Dataset on Kaggle.

Following the SEMMA Methodology

1. **Sample**
   The dataset contained 150 observations and five analysis columns after the species label was added. The project used an 80/20 train-test split, with stratification to preserve the proportion of each species. A fixed random state of 42 was used for reproducibility.

   The split produced 120 training observations and 30 test observations. Each species contributed 40 training examples and 10 test examples.

2. **Explore**
   I inspected descriptive statistics and compared average feature values across the three species. The dataset had no missing values in the selected fields, and each species had an equal number of observations.

   The exploratory pairplot showed that setosa was relatively easy to distinguish using petal measurements. Versicolor and virginica showed some overlap, suggesting that these two species could be more difficult to classify consistently.

3. **Modify**
   The four numeric measurements were used as model inputs, and species was used as the target label. The training and test data were separated before model fitting.

   For K-nearest neighbors (KNN), the features were standardized using StandardScaler within a scikit-learn pipeline. This is important because KNN uses distances between observations, and features with different scales can otherwise have unequal influence.

   The decision tree used the same training and test split but did not require feature standardization for this experiment.

4. **Model**
   Two models were trained:
   - **K-nearest neighbors:** a pipeline with standard scaling and five neighbors.
   - **Decision tree:** a classifier with random_state=42.

   Using the same split made it possible to compare the two models under the same test conditions.

5. **Assess**
   Both models reached **93.33% accuracy**, correctly predicting 28 out of 30 test samples.

   For KNN, all setosa and versicolor samples were correct, but two virginica samples were mislabeled as versicolor, giving virginica **80% recall**.

   The decision tree also got all setosa right. It made one mistake for versicolor and one for virginica, giving both species a precision, recall, and **F1 of 0.90**.

   Even though the models had the same overall accuracy, they made different mistakes:
   - KNN struggled only with virginica
   - The decision tree split its errors across the two harder-to-separate species

Limitations

This was a small learning experiment using a well-known dataset. Because it used only one train-test split, the accuracy results are just one estimate and could change with a different split. It also doesn't show how well the models would work on new flower samples or measurements collected in other conditions.

Better evaluation would use cross-validation, repeated splits, tuning the model settings, and a clearer comparison of how simple or interpretable each model is.

Conclusion

SEMMA gave a clear structure for working with the Iris dataset: sample, explore, modify, model, and assess. Both KNN and the decision tree reached 93.33% accuracy but made different mistakes on versicolor and virginica.

The results show why accuracy alone isn't enough. Confusion matrices and class-level metrics help reveal which classes are harder to separate and how the models differ.

Project Artifacts

The notebook and report are available in the SEMMA subdirectory of my repository: https://github.com/Olivia883/Cmpe255-data-mining-process/tree/main.